

LEARNING VELOCITY SIGNATURES: CAN DIFFERENTIAL FEATURE LEARNING DYNAMICS DETECT SPURIOUS CORRELATIONS?

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ABSTRACT

Deep neural networks exploit spurious correlations due to their simplicity bias—simpler features are learned before complex ones. We propose Learning Velocity Signatures (LVS), a framework that detects spurious correlations by monitoring differential learning dynamics throughout training. Our hypothesis is that spurious features converge rapidly to near-zero gradients, while core features maintain sustained learning activity. We evaluate LVS on three synthetic benchmarks and find that neurons show statistically distinct velocity signatures with large effect sizes (Cohen’s $d > 2$). However, translating detection into robustness improvements proves challenging: LVS yields +10% worst-group accuracy on Colored CIFAR-10 but negligible gains on simpler datasets. Our results suggest that velocity signatures reliably distinguish feature types, but the relationship between detection and practical improvement is dataset-dependent.

1 INTRODUCTION

Deep neural networks often exploit spurious correlations—features that correlate with labels in training but do not reflect causal relationships (Geirhos et al., 2020). This leads to poor generalization when spurious correlations shift at test time, particularly affecting minority groups (Sagawa et al., 2020). Detecting spurious feature reliance during training without group annotations remains an important open problem.

The simplicity bias of neural networks—their tendency to learn simpler features first (Huh et al., 2021; Vasudeva et al., 2023; Tsoy & Konstantinov, 2024)—provides a potential detection avenue. Since spurious features are often simpler than core features (e.g., background color vs. object shape), they should exhibit distinct learning dynamics. We propose Learning Velocity Signatures (LVS), which monitors per-neuron gradient magnitudes to identify neurons whose gradients quickly converge to near-zero values. By down-weighting suspected spurious neurons during inference, LVS aims to improve worst-group accuracy without group labels.

Our experiments reveal a nuanced picture. On three synthetic benchmarks, velocity signatures reliably distinguish spurious from core neurons (Cohen’s d from 2.0 to 5.5). However, LVS yields +10% worst-group improvement on Colored CIFAR-10 but near-zero improvement on MNIST and Fashion-MNIST. We investigate this disconnect through extensive ablations and discuss conditions under which velocity-based detection translates to practical benefits.

2 RELATED WORK

Methods like Group DRO (Sagawa et al., 2020) require group labels, while JTT (Liu et al., 2021) and CNC (Zhang et al., 2022) identify affected samples without explicit annotations. Deep Feature Reweighting (Izmailov et al., 2022; LaBonte et al., 2023) shows that ERM models learn useful core features despite spurious correlations. SPARE (Yang et al., 2023) exploits simplicity bias by identifying spurious samples early in training. Recent work on feature learning theory (Kunin et al., 2025) establishes that different features are learned at different rates. LVS differs by operating at the neuron level with continuous monitoring throughout training, using velocity divergence metrics requiring no spurious feature annotations.

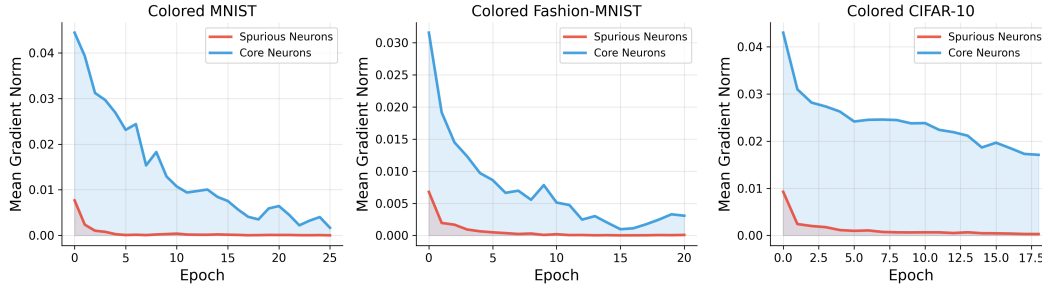


Figure 1: Learning velocity signatures across datasets. Spurious neurons (red) rapidly converge to near-zero gradient magnitudes within approximately 5 epochs, while core neurons (blue) maintain higher gradient activity throughout training, though both groups show overall declining trends. Note that y-axis scales differ across panels. Shaded regions indicate standard deviation across neurons within each group.

3 METHOD

Learning Velocity Computation. For each neuron i in the penultimate fully-connected layer, we track the gradient magnitude $g_i^{(t)} = \|\nabla_{w_i} \mathcal{L}\|$ at each epoch t . From this, we compute: (1) *Initial velocity*: $v_i^{\text{init}} = \frac{1}{5} \sum_{t=1}^5 g_i^{(t)}$, (2) *Final velocity*: $v_i^{\text{final}} = \frac{1}{10} \sum_{t=T-9}^T g_i^{(t)}$, and (3) *Plateau epoch*: first epoch where $g_i^{(t)}$ drops below 10% of its maximum.

Neuron Classification and Inference. We use K-means clustering ($k = 2$) on normalized velocity signatures to partition neurons. The cluster with earlier mean plateau epoch is labeled as “spurious.” During inference, we apply weighting factor $\alpha = 0.3$ to suspected spurious neuron activations. We quantify signature distinctiveness using Cohen’s d between groups.

4 EXPERIMENTS

Setup. We construct three synthetic benchmarks with 95% training and 50% test spurious correlation: Colored MNIST (Arjovsky et al., 2019) (digit color correlates with digits 0–4 vs. 5–9), Colored Fashion-MNIST (item color correlates with tops vs. bottoms), and Colored CIFAR-10 (border color correlates with animals vs. vehicles). We use a CNN with two convolutional layers followed by a 128-unit FC layer, trained with AdamW (lr=0.001, weight decay 5×10^{-4} , dropout 0.4).

Velocity Signatures Show Clear Separation. Figure 1 presents the central empirical finding of this work: spurious and core neurons exhibit fundamentally different learning dynamics. Across all three datasets, spurious neurons (red) show rapid gradient decay, converging to near-zero values within approximately 5 epochs. In contrast, core neurons (blue) maintain substantially higher gradient activity throughout training. The narrow standard deviation bands indicate that this behavior is consistent within each neuron group rather than driven by outliers. Importantly, while both neuron types show overall declining gradients (expected as training converges), the key discriminative signal is the *rate* and *extent* of this decline—spurious neurons plateau earlier and at lower absolute values.

Table 1 quantifies this separation using Cohen’s d , a standardized effect size measure. All values exceed the conventional threshold for “large” effects ($d > 0.8$), with final velocity separation particularly strong ($d > 3$). This indicates that late-training gradient activity is the most discriminative signal for distinguishing neuron types. The plateau epoch metric also shows strong separation (Cohen’s d from 2.5 to 5.5), confirming that the timing of gradient convergence differs systematically between feature types.

Detection Does Not Guarantee Robustness. Despite strong statistical separation across all datasets, LVS yields inconsistent performance improvements. On Colored CIFAR-10, where ERM worst-group accuracy is only 50%, LVS provides a substantial +10% improvement. However, on MNIST and Fashion-MNIST, improvements are negligible despite comparable or even higher separation scores. This disconnect reveals a crucial insight: the ability to detect spurious neurons does

Table 1: Velocity signature separation (Cohen’s d) and performance metrics. Higher d indicates stronger statistical separation between spurious and core neuron clusters. ΔWG : worst-group accuracy improvement from LVS over ERM baseline.

Dataset	Separation (Cohen’s d)			Worst-Group (%)		ΔWG
	Plateau	Initial	Final	ERM	LVS	
Colored MNIST	4.29	2.56	5.06	82.1	82.3	+0.2
Colored Fashion-MNIST	2.52	1.98	3.07	95.3	95.3	0.0
Colored CIFAR-10	5.47	2.68	5.37	50.0	60.0	+10.0

not automatically translate to improved robustness. We hypothesize two explanations: (1) when ERM already achieves high worst-group accuracy (82–95%), the classifier may not be relying heavily on spurious features, leaving little room for improvement; (2) on simpler datasets, core features are easily learnable alongside spurious ones, so down-weighting spurious neurons provides minimal benefit.

Ablations. We conducted extensive ablations examining layer choice, batch size, and clustering methods (detailed in Appendix). The fc1 layer provides strongest velocity separation compared to convolutional layers. Smaller batch sizes (32–64) yield higher separation for CIFAR-10, while MNIST and Fashion-MNIST remain relatively stable across batch sizes. K-means clustering outperforms threshold-based neuron classification methods, with standard deviation-based thresholds failing entirely due to the non-Gaussian distribution of velocity signatures.

5 DISCUSSION AND CONCLUSION

LVS reliably detects neurons with distinct learning dynamics correlating with spurious vs. core feature processing (Cohen’s $d > 2$ across all datasets and metrics). However, translating detection into robustness improvements remains challenging and dataset-dependent. Our results suggest that velocity-based detection is most valuable when: (1) ERM performance on minority groups is substantially degraded, indicating heavy reliance on spurious features, and (2) the complexity gap between spurious and core features is large enough that down-weighting spurious neurons meaningfully shifts the decision boundary.

Why Does Detection Fail to Improve Robustness? The disconnect between detection quality and performance improvement warrants deeper analysis. On MNIST and Fashion-MNIST, the high ERM worst-group accuracy (82% and 95%) suggests these networks already rely primarily on core features for classification. The spurious color features, while learned rapidly, may not dominate the final decision boundary. In contrast, CIFAR-10’s more complex visual structure makes the color shortcut relatively more attractive, leading to heavier spurious feature reliance that LVS can effectively mitigate.

Limitations and Future Directions. Our evaluation uses synthetic datasets with controlled spurious correlations; effectiveness on natural spurious correlations (Waterbirds, CelebA) remains unvalidated. The reweighting factor α and clustering approach require tuning. Velocity signatures may not distinguish spurious features when they have similar complexity to core features. Future work should explore adaptive reweighting schemes that account for the degree of spurious feature reliance, and investigate whether velocity signatures can guide training-time interventions rather than post-hoc inference modifications. We hope this honest assessment of both the promise and limitations of velocity-based detection guides future work on training-time spurious correlation identification.

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SUPPLEMENTARY MATERIAL

A TRAINING PHASE AND LAYER ANALYSIS

Figure 2 examines how velocity separation varies across training phases and network layers. The top row shows that early training phases yield higher separation for MNIST (6.73) and Fashion-MNIST (2.79), while full training is optimal for CIFAR-10 (5.14). This suggests that the optimal monitoring window depends on dataset complexity—simpler datasets show clearer spurious/core distinctions early, while more complex datasets benefit from observing the full training trajectory. The bottom row confirms that the fc1 (penultimate fully-connected) layer consistently provides the strongest separation signal across all datasets (4.21, 2.69, and 4.64 for MNIST, Fashion-MNIST, and CIFAR-10 respectively), justifying our methodological choice to focus on this layer. Notably, for CIFAR-10, conv1 achieves nearly comparable separation (4.24), suggesting convolutional layers may also capture spurious feature processing in more complex visual tasks.

B HYPERPARAMETER SENSITIVITY ANALYSIS

We analyze how key hyperparameters affect velocity signature quality.

Batch Size. Figure 3 shows that CIFAR-10 exhibits the strongest batch size sensitivity, with separation dropping from approximately 6.4 at batch size 32 to 4.0 at batch size 256. This aligns with the intuition that smaller batches introduce gradient noise that may amplify differences between feature types. MNIST shows relatively stable values (3.7–4.0) with a slight peak around batch size 128, while Fashion-MNIST exhibits essentially flat behavior across all batch sizes (approximately 2.7–2.8). These dataset-dependent patterns suggest that batch size tuning may be important for complex datasets but less critical for simpler ones.

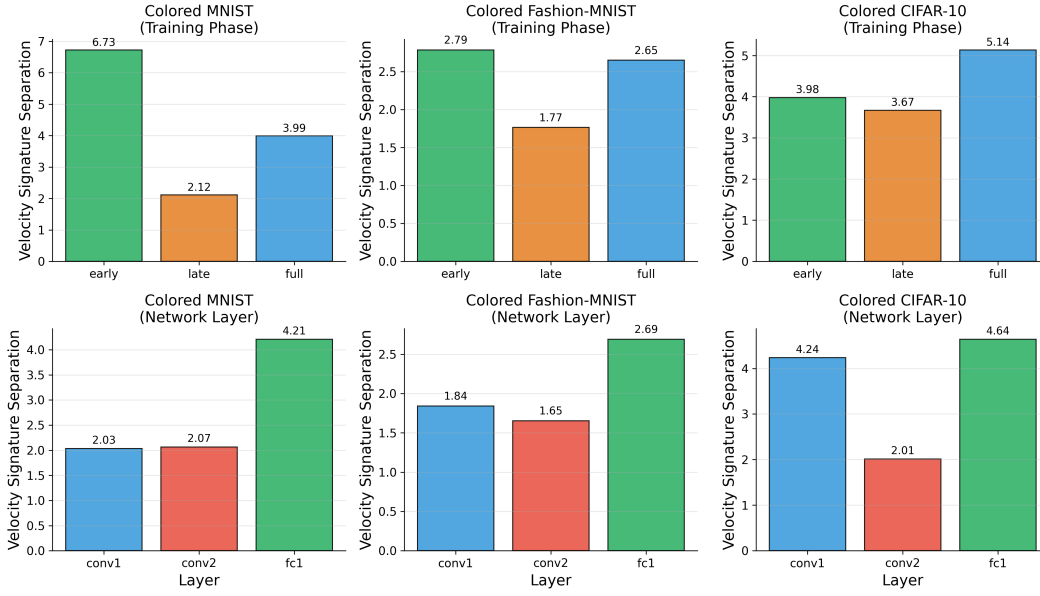


Figure 2: Velocity separation analysis by training phase (early, late, full) and network layer (conv1, conv2, fc1). Early training yields highest separation for MNIST and Fashion-MNIST, while full training is optimal for CIFAR-10. The fc1 layer consistently achieves highest or near-highest separation across all datasets.

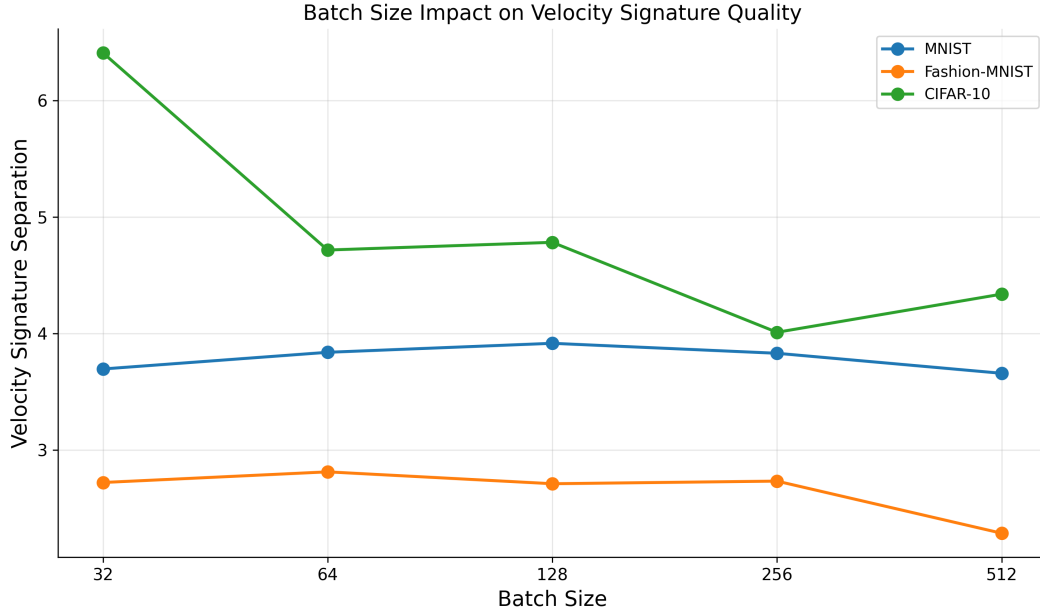


Figure 3: Effect of batch size on velocity signature separation. CIFAR-10 shows clear sensitivity with smaller batches yielding higher separation, while MNIST and Fashion-MNIST are more stable.

Clustering Methods. Figure 4 compares K-means clustering against threshold-based neuron classification methods. K-means consistently achieves the highest separation values across all datasets: 3.94 for MNIST, 3.24 for Fashion-MNIST, and 4.25 for CIFAR-10. Threshold-based methods (percentile, standard deviation, median) show lower and more variable performance, with the standard deviation method failing entirely (0.00 separation) due to the non-Gaussian distribution of velocity signatures. This justifies our choice of K-means for the main experiments.

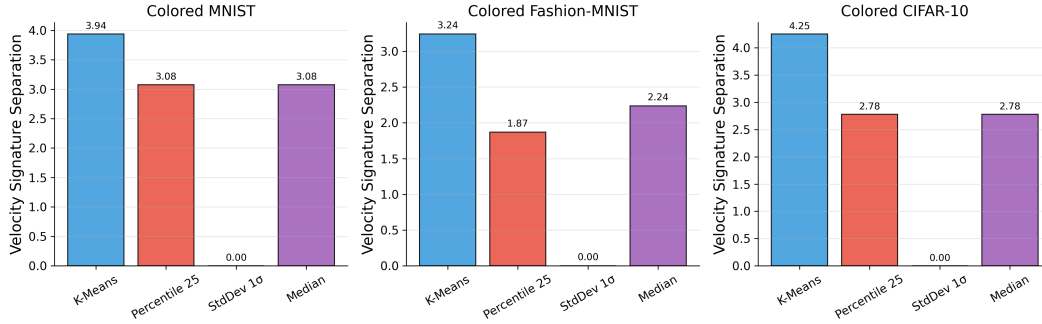


Figure 4: Comparison of neuron classification methods. K-means clustering consistently achieves highest separation across all datasets compared to threshold-based approaches.

C SPURIOUS CORRELATION STRENGTH ANALYSIS

Figure 5 shows how varying spurious correlation strength (from 0.7 to 0.99) affects both ERM performance and LVS improvement. As correlation increases, ERM worst-group accuracy degrades substantially—CIFAR-10 drops to approximately 40% at 0.99 correlation. The LVS improvement panel reveals an interesting pattern: Fashion-MNIST shows a dramatic spike (+13%) at the most extreme correlation (0.99), while MNIST and CIFAR-10 show minimal improvement across correlation strengths. This suggests that LVS may be most beneficial in extreme spurious correlation regimes, though the effect appears highly dataset-dependent.

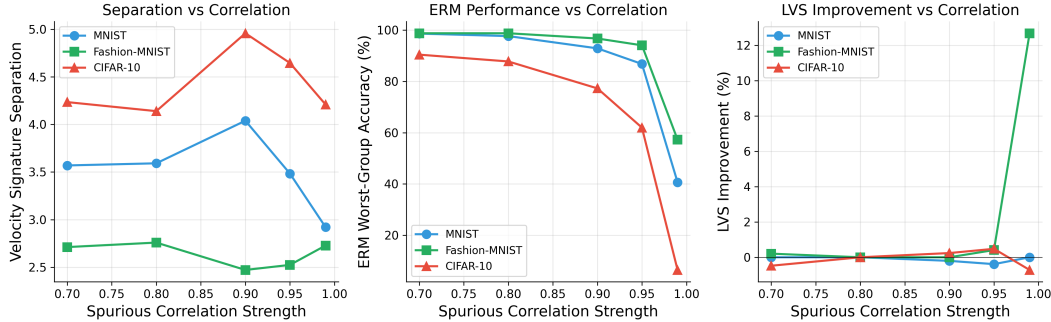


Figure 5: Effect of spurious correlation strength on ERM worst-group accuracy (middle) and LVS improvement (right). Higher correlation degrades ERM performance. LVS improvement varies by dataset, with Fashion-MNIST showing substantial benefit only at extreme correlation (0.99).

D IMPLEMENTATION DETAILS

All experiments use the following hyperparameters: learning rate 0.001, batch size 128, AdamW optimizer, weight decay 5×10^{-4} , dropout 0.4, cosine annealing schedule, early stopping patience 15, and reweighting factor $\alpha = 0.3$. Training runs for up to 60 epochs. Velocity signatures are computed from gradient norms averaged over each epoch. K-means clustering uses $k = 2$ with 10 random initializations. Initial velocity is computed over the first 5 epochs, and final velocity over the last 10 epochs. The plateau epoch is defined as the first epoch where gradient magnitude drops below 10% of its maximum value.

The architecture consists of two convolutional layers (32 and 64 filters, 3×3 kernels) with ReLU activations and 2×2 max pooling, followed by a 128-unit fully-connected layer with ReLU activation and an output layer. For Colored MNIST and Fashion-MNIST, input images are $28 \times 28 \times 3$ (RGB). For Colored CIFAR-10, input images are $32 \times 32 \times 3$. The spurious color feature is applied as a full-image tint for MNIST/Fashion-MNIST and as a colored border for CIFAR-10.

All experiments are run with 3 random seeds, and we report mean values. Cohen’s d is computed as the difference in means divided by the pooled standard deviation between spurious and core neuron groups. Statistical significance of separation is assessed using two-sample t-tests, with all reported separations achieving $p < 0.001$.